stat 578 scribing

Scribe: Haotian Ju Instructor: Joshua Agterberg

November 2025

1 Introduction

argmin $\hat{A} = 1/4 \sum_{(i,j)\in\Omega}(M_{i,j} - Z_{i,j}) + \lambda||Z||_k$ = argmin $1/4 \sum_{(i,j)\in\Omega}(M_{i,j} - (XY^T)_{i,j}) + \lambda/2(||X||_k + ||Y||_k)$
$\hat{M} = \hat{u}(\Sigma - \lambda I)_+ \hat{V}^T$ is biased
Dibiased$(\hat{M}) = \hat{M}^{HT}$. By Weyl's inequality: $\sigma\sqrt{n} \leq \lambda \leq \lambda_r(M_0) - c\sigma\sqrt{n}$
$Z - 1/pP_\Omega(Z - M) = (1/p(M_0)) + (1/pP_\Omega(E)) + (Z - 1/pP_\Omega(M))$, where the first term is approximately equal to M_0, the second term has mean 0, and the last term is small.

$$\mathbf{X}^d = \mathbf{X}\left(I_r + p^{-1}\lambda\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\right)^{1/2},$$

$$\mathbf{Y}^d = \mathbf{Y}\left(I_r + p^{-1}\lambda\left(\mathbf{Y}^\top\mathbf{Y}\right)^{-1}\right)^{1/2},$$

Under some conditions, we have:$\mathbf{X}^d\mathbf{Y}^d = \mathbf{M}^d$

Theorem: (Chen etal, PNAS, 2018) We define:

$$H^d = \arg\min_{R\in\mathcal{O}^{r\times r}} \left\|X^dR - X^\star\right\|_F^2 + \left\|Y^dR - Y^\star\right\|_F^2.$$

Suppose that the sample complexity meets $n^2p \geq C\kappa^8\mu^3r^2n\log^3 n$ for some sufficiently large constant $C > 0$ and the noise obeys $\dfrac{\sigma}{\sigma_{\min}} \leq c\sqrt{\dfrac{p}{\kappa^8\mu rn\log^2 n}}$ for some sufficiently small constant $c > 0$, then one can write:

$$X^dH^d - X^\star = Z_X + \Psi_X$$

and

$$Y^dH^d - Y^\star = Z_Y + \Psi_Y$$

Here, the rows of $Z_X \in \mathbb{R}^{n\times r}$ (resp. $Z_Y \in \mathbb{R}^{n\times r}$) are independent and obey:

$$Z_X^\top e_j \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{\sigma^2}{p}(\Sigma^\star)^{-1}\right), \qquad 1 \leq j \leq n$$

and

$$Z_Y^\top e_j \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{\sigma^2}{p}(\Sigma^\star)^{-1}\right), \qquad 1 \le j \le n$$

Also, we have:

$$\max\left\{\|\Psi_X\|_{2,\infty},\ \|\Psi_Y\|_{2,\infty}\right\} = o\left(\frac{\sigma\sqrt{r}}{\sqrt{p\sigma_{\max}}}\right).$$

Theorem2: (Chen etal, PNAS, 2018) For each $1 \le i, j \le n$, define the variance $v_{ij}^\star$ as:

$$v_{ij}^\star \triangleq \frac{\sigma^2}{p}\left(\|U_i^\star\|_2^2 + \|V_j^\star\|_2^2\right),$$

We assume that:

$$np \gtrsim \kappa^8\mu^3 r^3 \log^3 n, \qquad \sigma\sqrt{\frac{\kappa^8\mu r n \log^2 n}{p}} \lesssim \sigma_{\min},$$

and:

$$\|U_i^\star\|_2 + \|V_j^\star\|_2 \gtrsim \sqrt{\frac{r}{n}}\frac{\sigma}{\sigma_{\min}}\sqrt{\frac{\kappa^6\mu^2 r n \log^3 n}{p}},$$

Then:

$$M_{ij}^d - M_{ij}^\star = g_{ij} + \Delta_{ij},$$

$$g_{ij} \sim \mathcal{N}(0, v_{ij}^\star), \qquad |\Delta_{ij}| = o\left(\sqrt{v_{ij}^\star}\right).$$

Proportional Asymptotics (Traditional Statistics useless in high dimensional setting) $\lambda_r/(\sigma\sqrt{n}) \asymp 1$

$$Y_i = X_i^\top\beta^\star + \varepsilon_i, \qquad X_i \sim \mathcal{N}\left(0, \frac{1}{n}I_p\right), \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \ldots, n.$$

Let $X \in \mathbb{R}^{n\times p}$ be the design matrix with rows $X_i^\top$, and $y = (Y_1, \ldots, Y_n)^\top$.

$$\hat{\beta}_{\text{ridge}}(\lambda) = \arg\min_{\beta\in\mathbb{R}^p}\left\{\frac{1}{2n}\|y - X\beta\|_2^2 + \frac{\lambda}{2}\|\beta\|_2^2\right\} = \left(X^\top X/n + \lambda I_p\right)^{-1}(X^\top y/n).$$

and:

$$\hat{\beta}_{\text{lasso}}(\lambda) = \arg\min_{\beta\in\mathbb{R}^p}\left\{\frac{1}{2n}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1\right\}.$$

Assume $n \propto p \propto s$
$\eta(x, \lambda) = sign(x)(|x| - \lambda)_+$
$\tau^2 = \sigma^2 + 1/\delta E_{(\beta_0, G)}[(\eta(\beta_0 + \tau G, \tau\lambda/\gamma) - \beta_0)^2]$
$\gamma = \tau(1 - 1/\delta E_{(\beta_0, G)}[\eta(\beta_0 + \tau G, \tau\lambda/\beta)])$

β i.i.d β_0 and G follows standard normal distribution

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \frac{1}{2n}\|Y - X\beta\|_2^2 + \frac{\lambda}{\sqrt{n}}\|\beta\|_1,$$

$$Y = X\beta + \varepsilon, \qquad y^\delta = \delta + \frac{\tau}{\sqrt{n}} g, \qquad \frac{n}{p} \to \delta \in (0, \infty).$$

$$\hat{\theta}(s, z) = \arg\min_{\theta} \frac{1}{2}\|y^\delta - \theta\|_2^2 + \frac{\lambda}{s\sqrt{n}}\|\theta\|_1,$$

Particular (s^*, z^*) will have

$$\hat{\beta} \overset{d}{\approx} \hat{\theta}(s^*, z^*).$$

The corresponding soft-thresholding form:

$$\hat{\theta}(\S, \tau) = \left(|y^\delta| + \tfrac{\lambda}{\S n}\right)_+ \operatorname{sign}(y^\delta).$$

System:

$$\begin{cases} \tau^2 = \sigma^2 + \mathbb{E}\left\|\hat{\theta}(\S, \tau) - \theta\right\|_2^2, \\ \S = 1 - \dfrac{\sqrt{n}\,\mathbb{E}\langle \hat{\theta}(\S, \tau), g\rangle}{\tau}. \end{cases} \qquad \text{(solve for } (\tau^*, \S^*)\text{)}$$

Equivalently,

$$\tau^2 = \sigma^2 + \mathbb{E}\left\|\hat{\theta}(\S, \tau) - \theta\right\|_2^2, \qquad \S = 1 - \frac{\mathbb{E}\|\hat{\theta}(\S - \tau)\|_0}{n}.$$

For β_i i.i.d.:

$$\hat{\S} = 1 - \frac{p}{n}\,\mathbb{P}(\beta_i \neq 0), \qquad \mathbb{E}\left\|\hat{\theta} - \theta\right\|_2^2 \approx \mathbb{E}\left\|\hat{\beta} - \beta\right\|_2^2.$$

Thm 1. Suppose $\dfrac{s\log(p/\delta)}{n} \geq \Delta + L, \qquad \phi : \mathbb{R}^p \to \mathbb{R}$ is 1-Lipschitz.

$$\forall \varepsilon < L, \quad \mathbb{P}\Big(\exists \lambda \in [\lambda_{\min}, \lambda_{\max}] : \left|\phi(\hat{\beta}_\lambda^L) - \mathbb{E}[\phi(\hat{\theta}(\tau^*, s^*))]\right| > \varepsilon\Big) \leq \frac{c}{\epsilon^4}\exp(-cp\varepsilon^4).$$

Let

$$\Phi(G) := \min_{\mathbf{w}\in\mathcal{S}_w} \max_{\mathbf{u}\in\mathcal{S}_u} \mathbf{u}^\top G\mathbf{w} + \psi(\mathbf{w}, \mathbf{u}),$$

and

$$\phi(\mathbf{g}, \mathbf{h}) := \min_{\mathbf{w}\in\mathcal{S}_w} \max_{\mathbf{u}\in\mathcal{S}_u} \|\mathbf{w}\|\,\mathbf{g}^\top\mathbf{u} + \|\mathbf{u}\|\,\mathbf{h}^\top\mathbf{w} + \psi(\mathbf{w}, \mathbf{u}),$$

where $G \in \mathbb{R}^{m\times n}$, $\mathbf{g} \in \mathbb{R}^m$, $\mathbf{h} \in \mathbb{R}^n$, $\mathcal{S}_w \subset \mathbb{R}^n$, $\mathcal{S}_u \subset \mathbb{R}^m$, and $\psi : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$.

Theorem (CGMT). Assume that in the above display, $\mathcal{S}_w, \mathcal{S}_u$ are compact sets, ψ is continuous on $\mathcal{S}_w \times \mathcal{S}_u$, and $G, \mathbf{g}, \mathbf{h}$ all have entries i.i.d. standard normal. Then the following hold:

$$\text{(i)} \quad \mathbb{P}\big(\Phi(G) < c\big) \le 2\,\mathbb{P}\big(\phi(\mathbf{g},\mathbf{h}) \le c\big), \quad \forall c \in \mathbb{R}.$$

If, in addition, $\mathcal{S}_w, \mathcal{S}_u$ are convex and ψ is convex–concave on $\mathcal{S}_w \times \mathcal{S}_u$, then

$$\text{(ii)} \quad \mathbb{P}\big(\Phi(G) > c\big) \le 2\,\mathbb{P}\big(\phi(\mathbf{g},\mathbf{h}) \ge c\big), \quad \forall c \in \mathbb{R}.$$